SET 26

Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

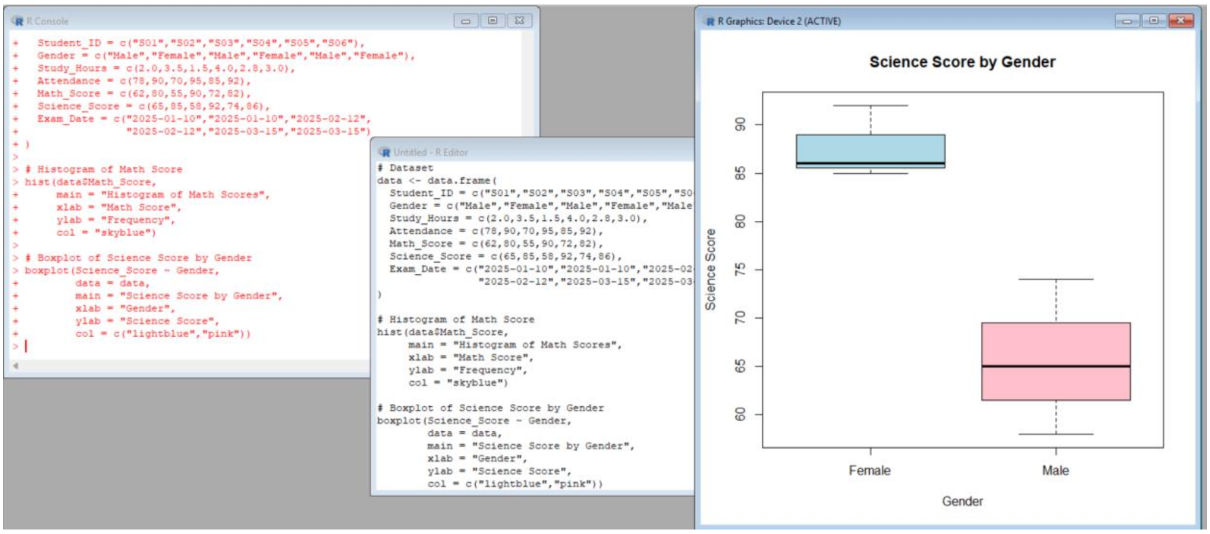
1.Import the dataset **in R**. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

A. CODE

```
data <- data.frame(
  Student_ID = c("S01","S02","S03","S04","S05","S06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Study_Hours = c(2.0,3.5,1.5,4.0,2.8,3.0),
  Attendance = c(78,90,70,95,85,92),
  Math_Score = c(62,80,55,90,72,82),
  Science_Score = c(65,85,58,92,74,86),
  Exam_Date = c("2025-01-10","2025-01-10","2025-02-12",
                "2025-02-12","2025-03-15","2025-03-15")
)
hist(data$Math_Score,
     main = "Histogram of Math Scores",
     xlab = "Math Score",
     ylab = "Frequency",
     col = "skyblue")
boxplot(Science_Score ~ Gender,
        data = data,
```

```
        main = "Science Score by Gender",
        xlab = "Gender",
        ylab = "Science Score",
        col = c("lightblue","pink"))
```

OUTPUT:



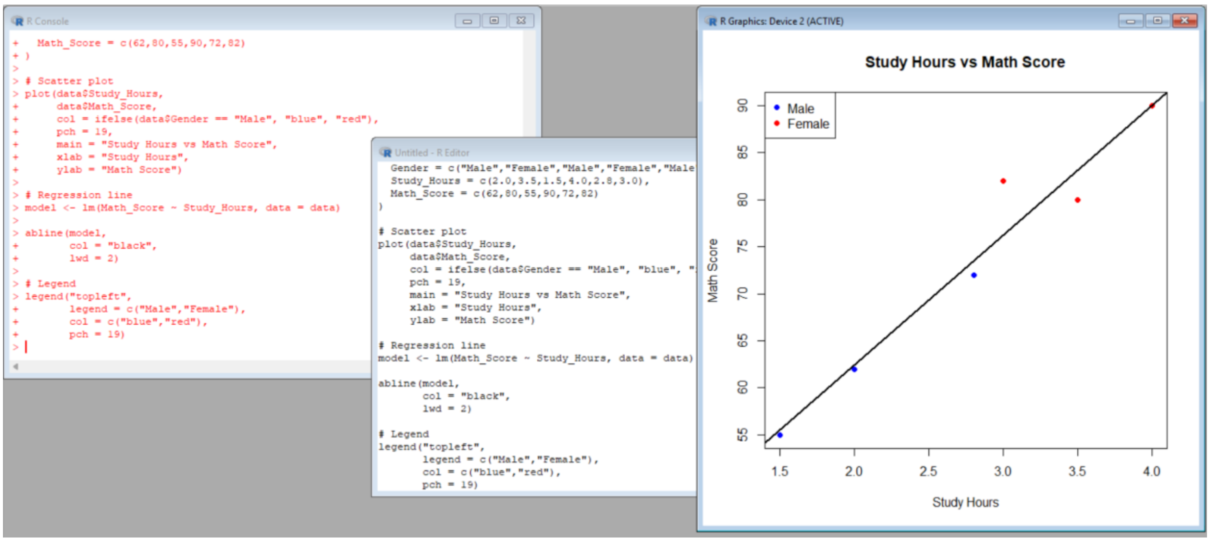
2.Create a scatter plot **using R programming** between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

A. CODE

```
data <- data.frame(
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Study_Hours = c(2.0,3.5,1.5,4.0,2.8,3.0),
  Math_Score = c(62,80,55,90,72,82)
)
plot(data$Study_Hours,
     data$Math_Score,
     col = ifelse(data$Gender == "Male", "blue", "red"),
     pch = 19,
     main = "Study Hours vs Math Score",
     xlab = "Study Hours",
     ylab = "Math Score")
model <- lm(Math_Score ~ Study_Hours, data = data)
```

```
abline(model,
       col = "black",
       lwd = 2)
legend("topleft",
       legend = c("Male","Female"),
       col = c("blue","red"),
       pch = 19)
```

OUTPUT:



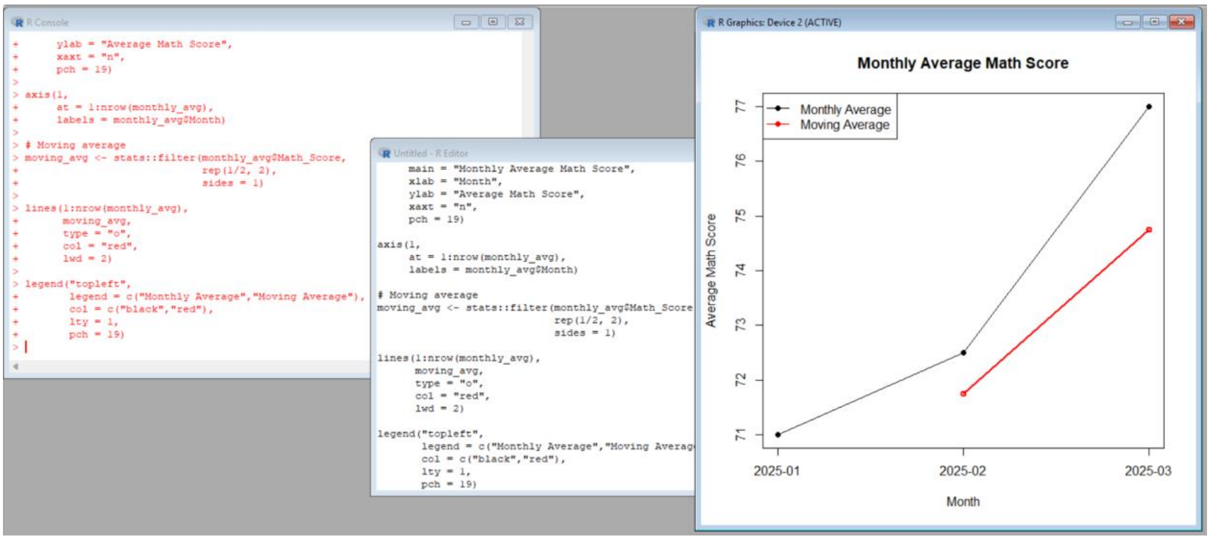
3.Using **R**, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

A. CODE

```
data <- data.frame(
  Exam_Date = c("2025-01-10","2025-01-10",
                "2025-02-12","2025-02-12",
                "2025-03-15","2025-03-15"),
  Math_Score = c(62,80,55,90,72,82)
)
data$Exam_Date <- as.Date(data$Exam_Date)
data$Month <- format(data$Exam_Date, "%Y-%m")
monthly_avg <- aggregate(Math_Score ~ Month,
                         data = data,
```

```
                    FUN = mean)
print(monthly_avg)
plot(1:nrow(monthly_avg),
     monthly_avg$Math_Score,
     type = "o",
     main = "Monthly Average Math Score",
     xlab = "Month",
     ylab = "Average Math Score",
     xaxt = "n",
     pch = 19)
axis(1,
     at = 1:nrow(monthly_avg),
     labels = monthly_avg$Month)
moving_avg <- stats::filter(monthly_avg$Math_Score,
                     rep(1/2, 2),
                     sides = 1)
lines(1:nrow(monthly_avg),
      moving_avg,
      type = "o",
      col = "red",
      lwd = 2)
legend("topleft",
       legend = c("Monthly Average","Moving Average"),
       col = c("black","red"),
       lty = 1,
       pch = 19)
```

OUTPUT



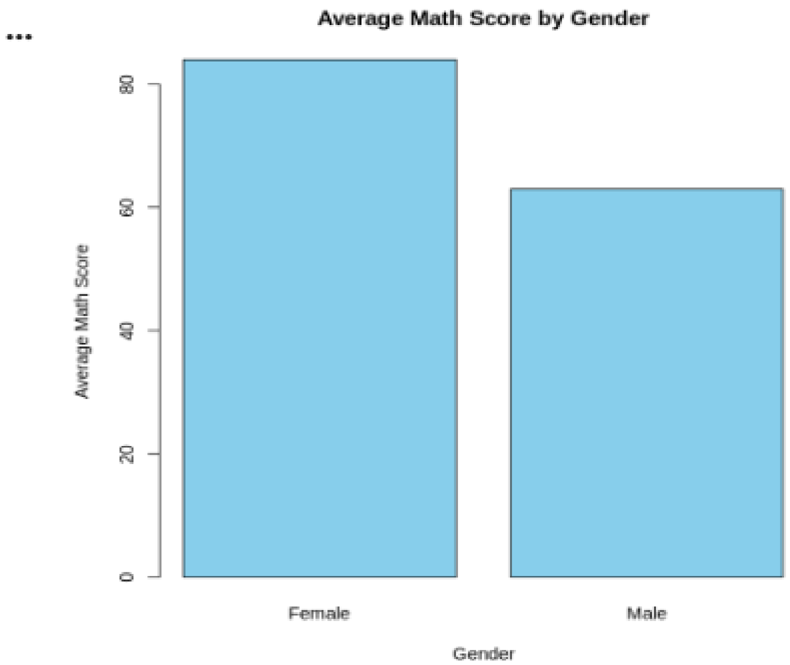
4.Import Student_Mini_Data.csv into **Tableau**. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

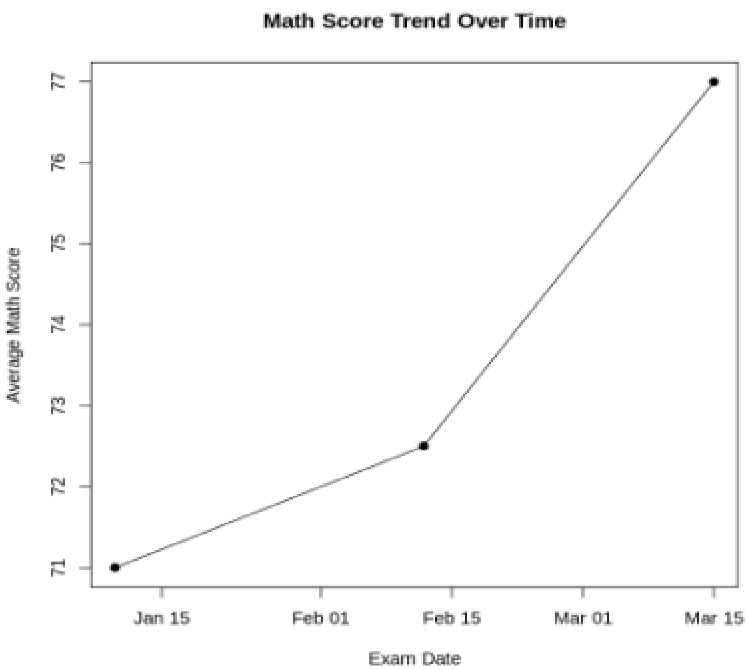
A. CODE

```
Student_ID <- c("S01","S02","S03","S04","S05","S06")

Gender <- c("Male","Female","Male","Female","Male","Female")

Math_Score <- c(62,80,55,90,72,82)

Exam_Date <- c("2025-01-10","2025-01-10",

               "2025-02-12","2025-02-12",

               "2025-03-15","2025-03-15"

average_math <- aggregate(Math_Score ~ Gender,

                          FUN = mean)

barplot(average_math$Math_Score,

        names.arg = average_math$Gender,

        main = "Average Math Score by Gender",

        xlab = "Gender",

        ylab = "Average Math Score",

        col = "skyblue")

Exam_Date <- as.Date(Exam_Date)

monthly <- aggregate(Math_Score ~ Exam_Date,

                     FUN = mean)

plot(monthly$Exam_Date,
```

```
    monthly$Math_Score,
    type = "o",
    pch = 19,
    main = "Math Score Trend Over Time",
    xlab = "Exam Date",
    ylab = "Average Math Score")
```

OUTPUT:





SET 27.

Patient Health Risk Analysis

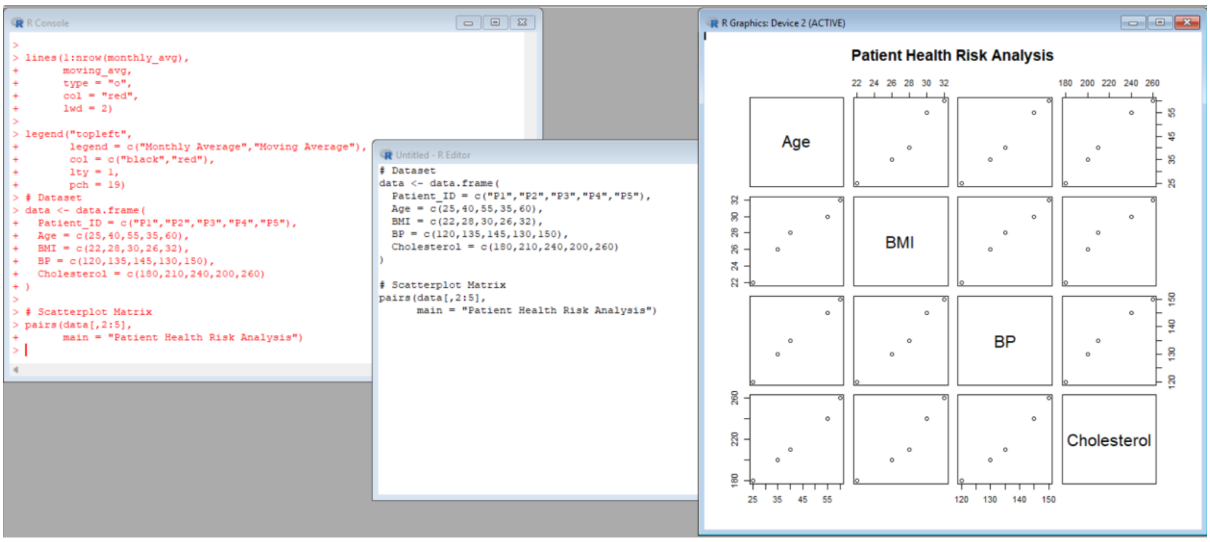
Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol.Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

A. CODE

```
data <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)
pairs(data[,2:5],
      main = "Patient Health Risk Analysis")
```

OUTPUT

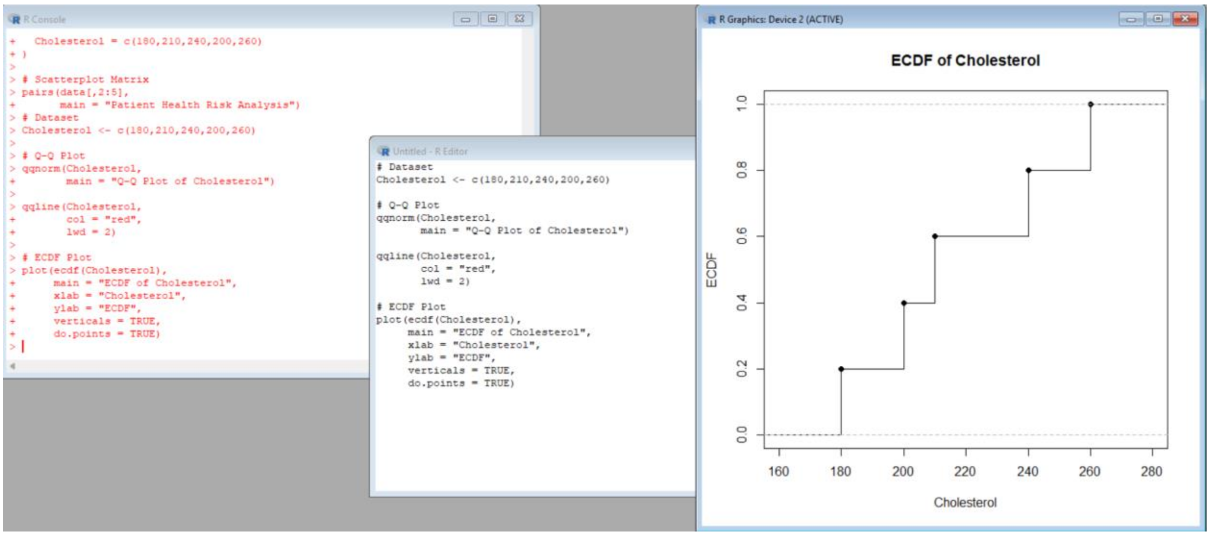


2.Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

A. CODE

```
Cholesterol <- c(180,210,240,200,260)
qqnorm(Cholesterol,
       main = "Q-Q Plot of Cholesterol")
qqline(Cholesterol,
       col = "red",
       lwd = 2)
plot(ecdf(Cholesterol),
     main = "ECDF of Cholesterol",
     xlab = "Cholesterol",
     ylab = "ECDF",
     verticals = TRUE,
     do.points = TRUE)
```

OUTPUT



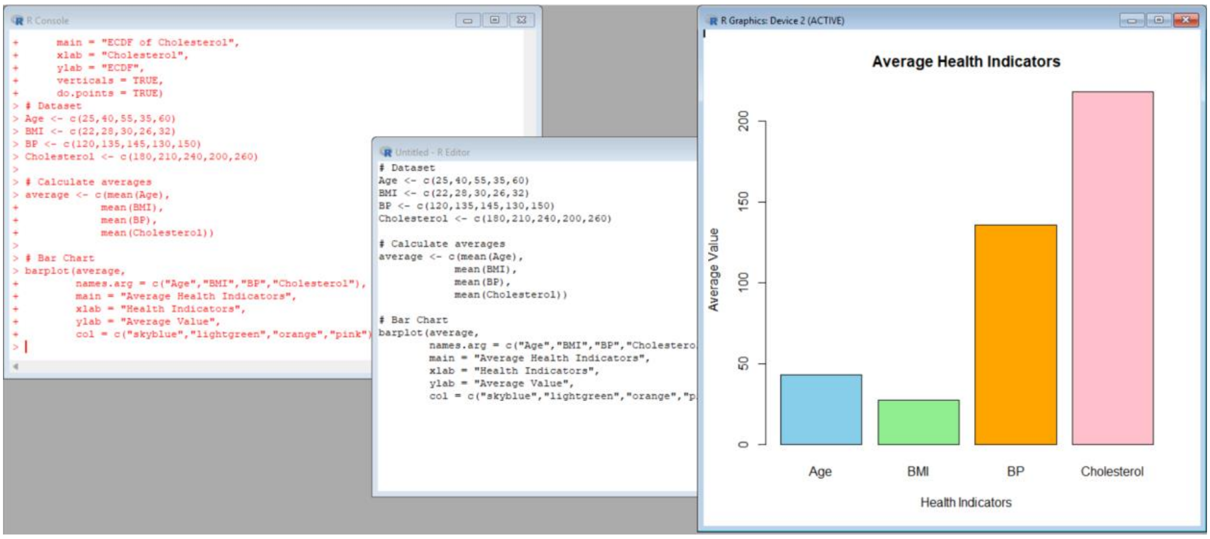
3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

A. CODE

```
Age <- c(25,40,55,35,60)
BMI <- c(22,28,30,26,32)
```

```
BP <- c(120,135,145,130,150)

Cholesterol <- c(180,210,240,200,260)

average <- c(mean(Age),

           mean(BMI),

           mean(BP),

           mean(Cholesterol))

barplot(average,

        names.arg = c("Age","BMI","BP","Cholesterol"),

        main = "Average Health Indicators",

        xlab = "Health Indicators",

        ylab = "Average Value",

        col = c("skyblue","lightgreen","orange","pink"))
```

OUTPUT



4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

A. CODE

```
data <- data.frame(

  Patient_ID = c("P1","P2","P3","P4","P5"),

  Age = c(25,40,55,35,60),

  BMI = c(22,28,30,26,32),
```

```
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)
data$Risk_Score <- data$BMI + data$BP/10 + data$Cholesterol/10
print(data)
write.csv(data,
          "Patient_Health_Risk.csv",
          row.names = FALSE)
```

OUTPUT

```
    Patient_ID Age BMI  BP Cholesterol Risk_Score
  1         P1  25  22 120         180       52.0
  2         P2  40  28 135         210       62.5
  3         P3  55  30 145         240       68.5
  4         P4  35  26 130         200       59.0
  5         P5  60  32 150         260       73.0
```

SET 28. Vehicle Performance Analysis

Dataset:

Vehicle_ID	**Engine_Size (L)**	**Horsepower**	**Fuel_Efficiency (km/l)**	**Top_Speed (km/h)**	**Safety_Rating**
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

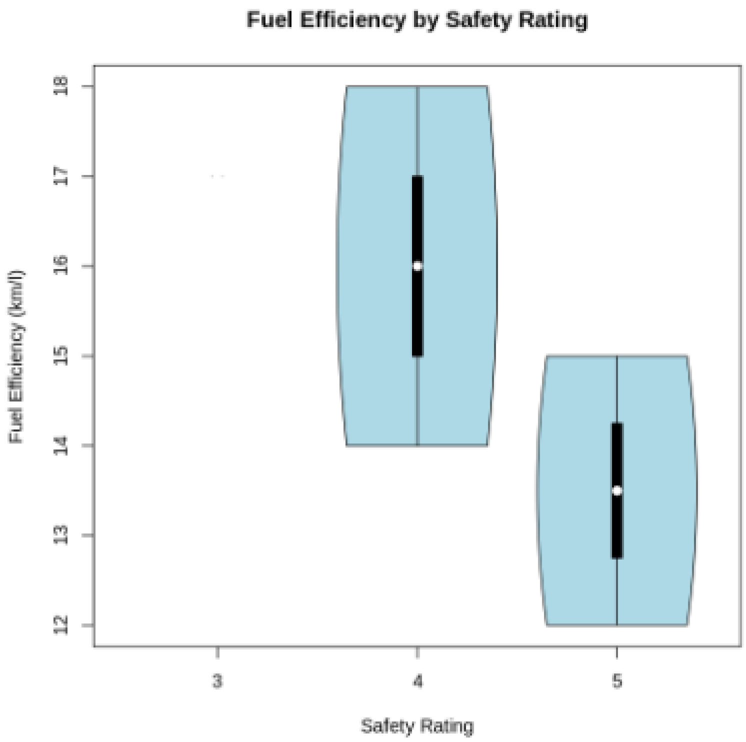
1.Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

CODE

```
data <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
```

```
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)
install.packages("vioplot")
library(vioplot)
vioplot(
  Fuel_Efficiency ~ Safety_Rating,
  data = data,
  main = "Fuel Efficiency by Safety Rating",
  xlab = "Safety Rating",
  ylab = "Fuel Efficiency (km/l)",
  col = "lightblue"
)
```

OUTPUT

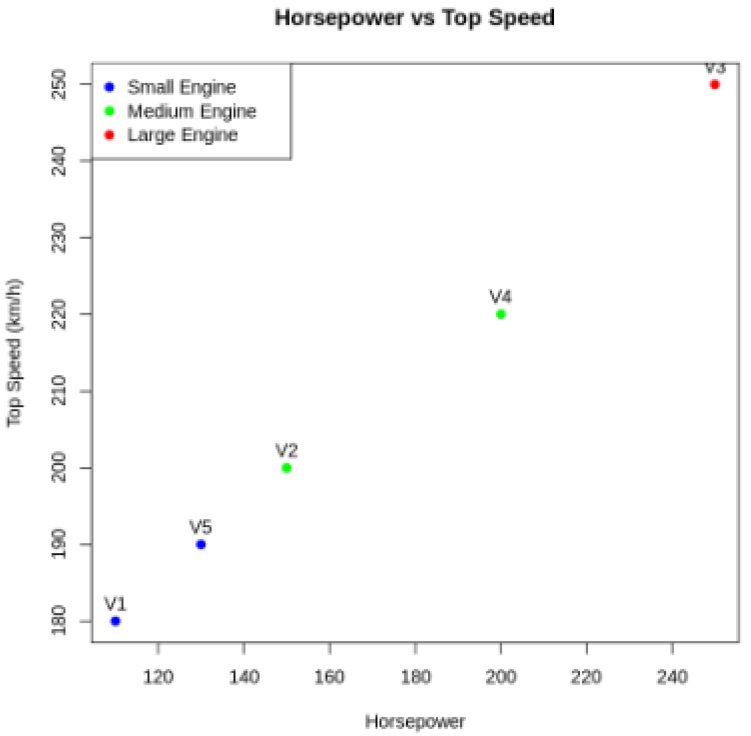


2.Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

CODE

```
data <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)
colors <- ifelse(data$Engine_Size <= 1.8, "blue",
                 ifelse(data$Engine_Size <= 2.5, "green", "red"))
plot(data$Horsepower,
     data$Top_Speed,
     main = "Horsepower vs Top Speed",
     xlab = "Horsepower",
     ylab = "Top Speed (km/h)",
     pch = 19,
     col = colors)
text(data$Horsepower,
     data$Top_Speed,
     labels = data$Vehicle_ID,
     pos = 3)
legend("topleft",
       legend = c("Small Engine","Medium Engine","Large Engine"),
       col = c("blue","green","red"),
       pch = 19)
```

OUTPUT



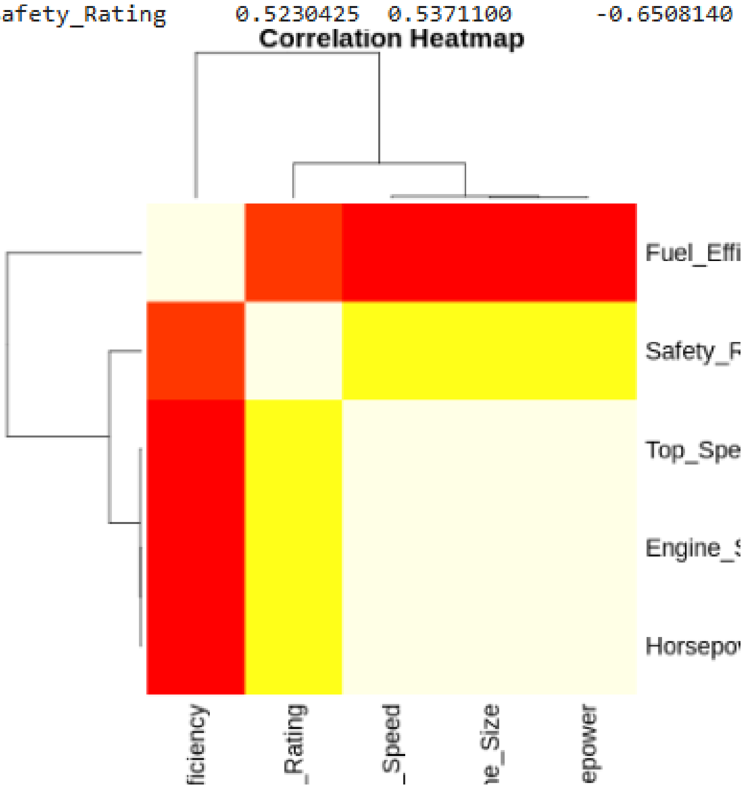
3.Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

CODE

```
data <- data.frame(
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)
correlation <- cor(data)
print(correlation)
heatmap(correlation,
        main = "Correlation Heatmap",
        symm = TRUE,
        col = heat.colors(20))
```

OUTPUT

```
...                Engine_Size Horsepower Fuel_Efficiency  Top_Speed Safety_Rating
    Engine_Size       1.0000000  0.9980902      -0.9799186  0.9947436     0.5230425
    Horsepower        0.9980902  1.0000000      -0.9743356  0.9970716     0.5371100
    Fuel_Efficiency  -0.9799186 -0.9743356       1.0000000 -0.9735918    -0.6508140
    Top_Speed         0.9947436  0.9970716      -0.9735918  1.0000000     0.5599496
    Safety_Rating     0.5230425  0.5371100      -0.6508140  0.5599496     1.0000000
```



4.In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

CODE

```
data <- data.frame(

  Vehicle_ID = c("V1","V2","V3","V4","V5"),

  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),

  Horsepower = c(110,150,250,200,130),

  Fuel_Efficiency = c(18,15,12,14,17),

  Top_Speed = c(180,200,250,220,190),

  Safety_Rating = c(4,5,5,4,3)

)

write.csv(data,

          "Vehicle_Performance.csv",
```

```
    row.names = FALSE)
print(data)
```

OUTPUT

```
  Vehicle_ID Engine_Size Horsepower Fuel_Efficiency Top_Speed Safety_Rating
1         V1         1.5        110              18       180             4
2         V2         2.0        150              15       200             5
3         V3         3.0        250              12       250             5
4         V4         2.5        200              14       220             4
5         V5         1.8        130              17       190             3
```

SET 29 .Student Academic Performance

Dataset:

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

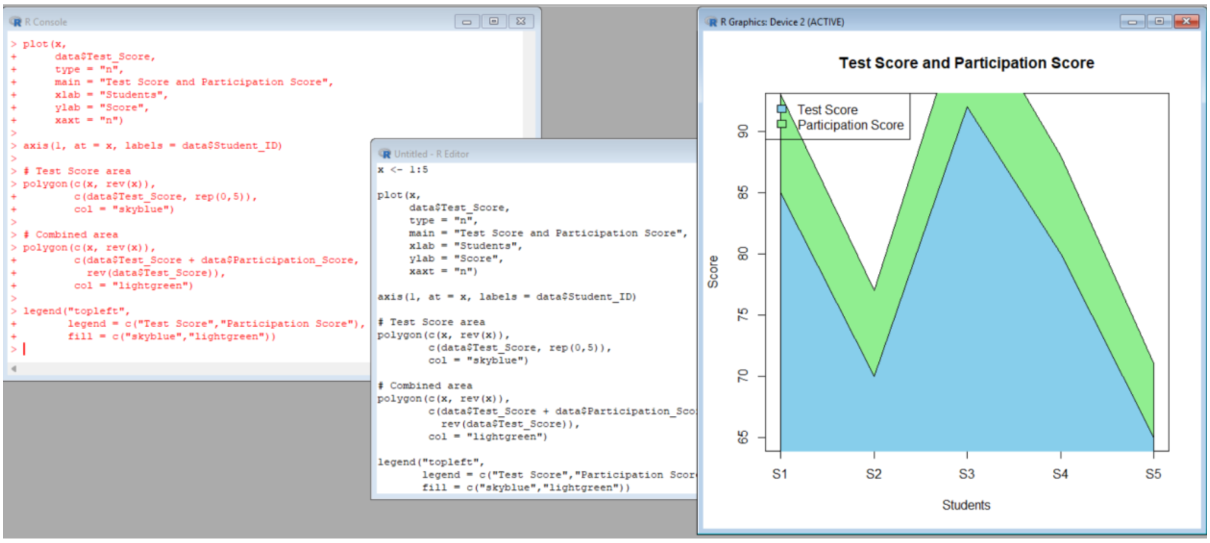
1.Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

CODE

```
data <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)
x <- 1:5
plot(x,
     data$Test_Score,
     type = "n",
```

```
     main = "Test Score and Participation Score",
     xlab = "Students",
     ylab = "Score",
     xaxt = "n")
axis(1, at = x, labels = data$Student_ID)
polygon(c(x, rev(x)),
        c(data$Test_Score, rep(0,5)),
        col = "skyblue")
polygon(c(x, rev(x)),
        c(data$Test_Score + data$Participation_Score,
          rev(data$Test_Score)),
        col = "lightgreen")
legend("topleft",
       legend = c("Test Score","Participation Score"),
       fill = c("skyblue","lightgreen"))
```

OUTPUT



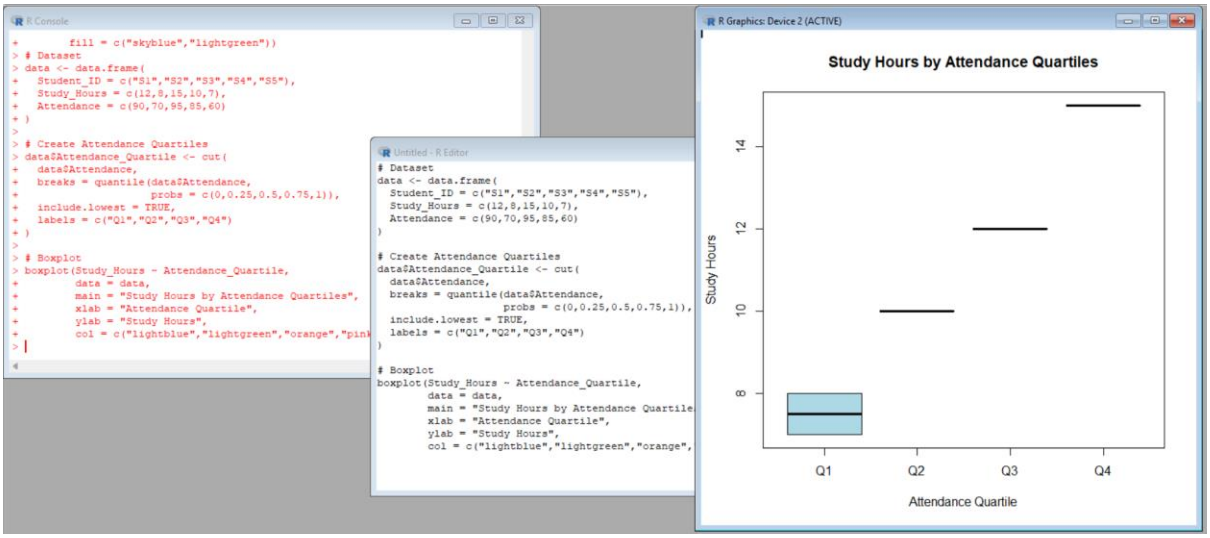
2.Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

CODE

```
data <- data.frame(
```

```
  Student_ID = c("S1","S2","S3","S4","S5"),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60)
)
data$Attendance_Quartile <- cut(
  data$Attendance,
  breaks = quantile(data$Attendance,
                    probs = c(0,0.25,0.5,0.75,1)),
  include.lowest = TRUE,
  labels = c("Q1","Q2","Q3","Q4")
)
boxplot(Study_Hours ~ Attendance_Quartile,
        data = data,
        main = "Study Hours by Attendance Quartiles",
        xlab = "Attendance Quartile",
        ylab = "Study Hours",
        col = c("lightblue","lightgreen","orange","pink"))
```

OUTPUT



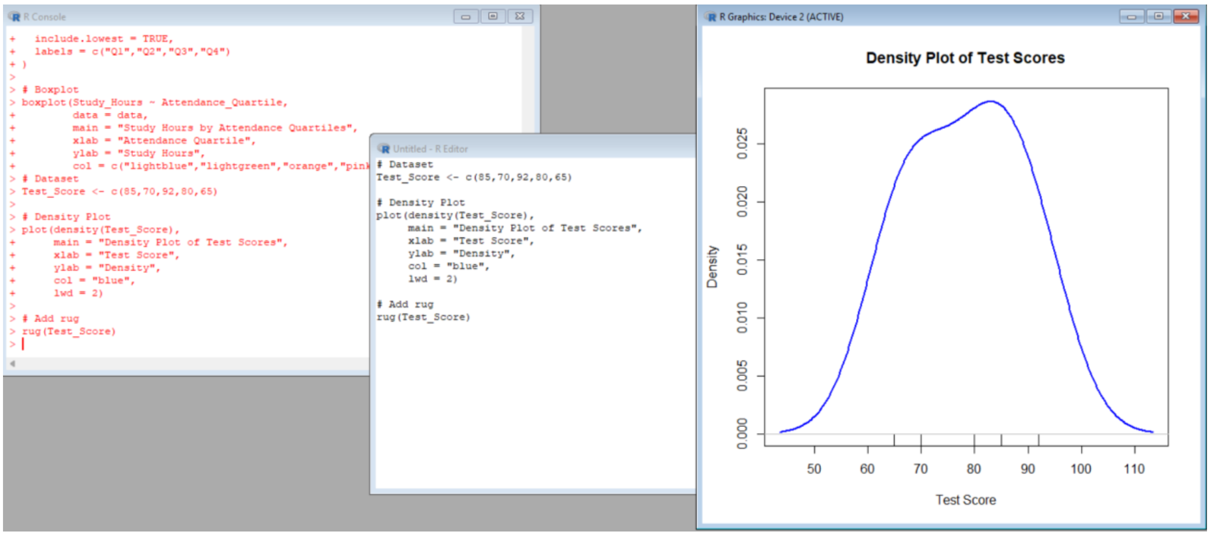
3.Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

CODE

```
Test_Score <- c(85,70,92,80,65)
```

```
plot(density(Test_Score),
     main = "Density Plot of Test Scores",
     xlab = "Test Score",
     ylab = "Density",
     col = "blue",
     lwd = 2)
rug(Test_Score)
```

OUTPUT



4.In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

CODE

```
data <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)
write.csv(data,
        "Student_Academic_Performance.csv",
        row.names = FALSE)
```

print(data)

OUTPUT

```
  Student_ID Age Study_Hours Attendance Test_Score Participation_Score
1         S1  19          12         90         85                   8
2         S2  21           8         70         70                   7
3         S3  20          15         95         92                   9
4         S4  22          10         85         80                   8
5         S5  23           7         60         65                   6
```

SET 30

Dataset: Mobile_App_Usage.csv

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

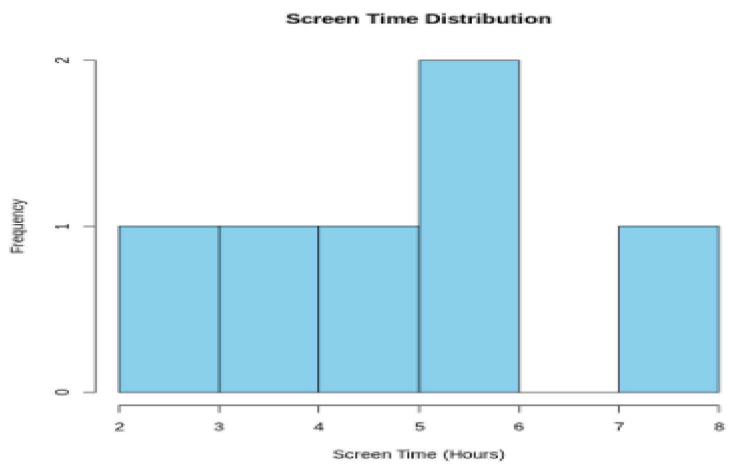
1.Import the dataset **in R**. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

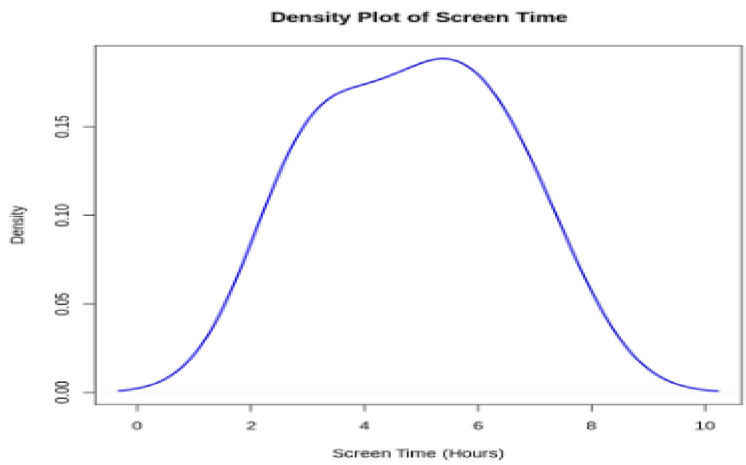
CODE

```
data <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = c("2025-01-08","2025-01-08",
               "2025-02-11","2025-02-11",
               "2025-03-14","2025-03-14")
```

```
)
hist(data$Screen_Time,
     main = "Screen Time Distribution",
     xlab = "Screen Time (Hours)",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")
plot(density(data$Screen_Time),
     main = "Density Plot of Screen Time",
     xlab = "Screen Time (Hours)",
     ylab = "Density",
     col = "blue",
     lwd = 2)
```

OUTPUT





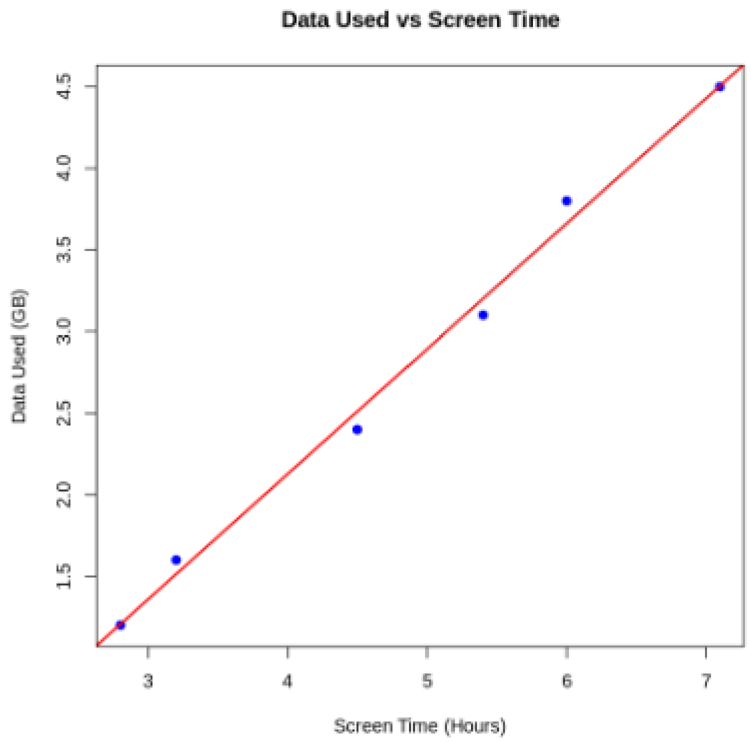
2.Create a scatter plot using **R programming** between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

CODE

```r
data <- data.frame(
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1)
)
plot(data$Screen_Time,
     data$Data_Used,
     main = "Data Used vs Screen Time",
     xlab = "Screen Time (Hours)",
     ylab = "Data Used (GB)",
     pch = 19,
     col = "blue")
correlation <- cor(data$Screen_Time,
                   data$Data_Used)
print(correlation)
model <- lm(Data_Used ~ Screen_Time,
            data = data)
abline(model,
       col = "red",
       lwd = 2)
```

OUTPUT

```
[1] 0.9969674
```



3.Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

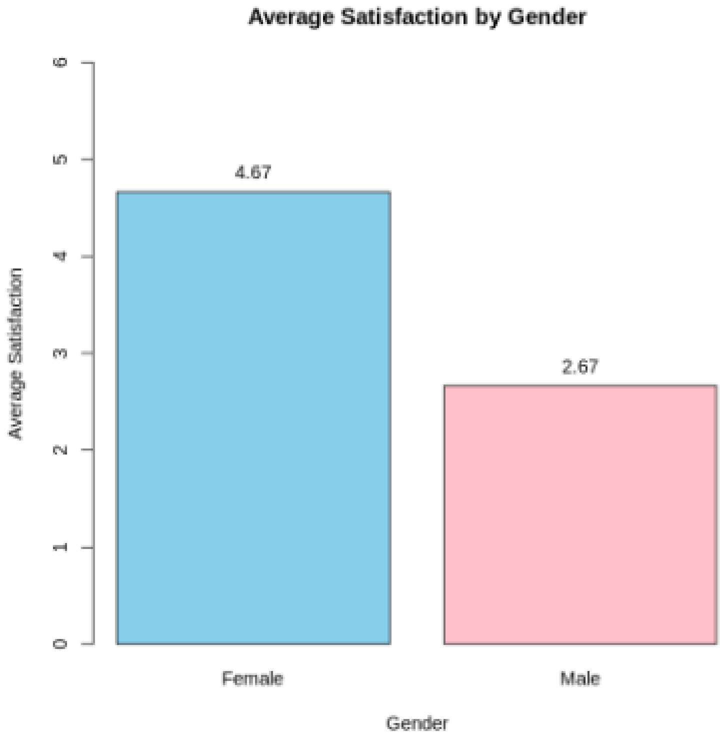
CODE

```
data <- data.frame(
  Gender = c("Male","Female","Male",
             "Female","Male","Female"),
  Satisfaction = c(3,5,3,5,2,4)
)
average <- aggregate(Satisfaction ~ Gender,
                     data = data,
                     FUN = mean)
print(average)
bars <- barplot(average$Satisfaction,
                names.arg = average$Gender,
                main = "Average Satisfaction by Gender",
                xlab = "Gender",
                ylab = "Average Satisfaction",
```

```
            col = c("skyblue","pink"),
            ylim = c(0,6))
text(bars,
     average$Satisfaction + 0.2,
     labels = round(average$Satisfaction,2))
```

OUTPUT

```
  Gender Satisfaction
1 Female     4.666667
2   Male     2.666667
```



4.Import Mobile_App_Usage.csv **into Tableau**. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.

CODE

```
data <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
```

```
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = c("2025-01-08","2025-01-08",
                 "2025-02-11","2025-02-11",
                 "2025-03-14","2025-03-14")
)
data$Usage_Date <- as.Date(data$Usage_Date)
write.csv(data,
          "Mobile_App_Usage.csv",
          row.names = FALSE)
print(data)
```

OUTPUT

```
  User_ID Gender Age Screen_Time App_Usage_Count Data_Used Satisfaction
1     U01   Male  20         4.5              18       2.4            3
2     U02 Female  22         6.0              25       3.8            5
3     U03   Male  19         3.2              12       1.6            3
4     U04 Female  21         7.1              30       4.5            5
5     U05   Male  23         2.8              10       1.2            2
6     U06 Female  20         5.4              22       3.1            4
  Usage_Date
1 2025-01-08
2 2025-01-08
3 2025-02-11
4 2025-02-11
5 2025-03-14
6 2025-03-14
```